

Project M2: Employee Attrition Analysis

Technical Workflow, Data Pipeline Summary & Executive Strategic Report

1. Executive Overview & Problem Statement

Business Context: XYZ Company employs approximately 4,000 workers with an annual attrition rate of 15%. Employee turnover causes severe project delays, client dissatisfaction, elevated recruitment costs, and continuous onboarding overheads.

Analytical Objective: The goal of Project M2 is to consolidate disparate HR datasets, calculate daily working hours, and perform quantitative drivers analysis to identify why employees leave (Attrition = Yes) versus why they stay (Attrition = No).

2. Created Files & Technical Pipeline Summary

To process and clean the raw data efficiently, a structured Python pipeline was built. Below is the comprehensive list of files created, along with their technical purpose:

File Name	Type	Tool / Method	Purpose & Description
data_merging.py	Python Script	Pandas (`pd.merge`)	Automates the merging of 3 survey datasets (`general_data.csv`, `employee_survey_data.csv`, `manager_survey_data.csv`) using `EmployeeID` as the primary key.
Master_Merged_Data.csv	Dataset (CSV)	Python Output	Consolidated master CSV containing demographics, survey feedback, role ratings, and manager assessments for all employees.
calculate_hours.py	Python Script	Pandas (`to_datetime`)	Processes multi-column timestamp logs (`in_time.csv` & `out_time.csv`), calculates precise daily working durations, and computes average daily working hours per employee.
Final_M2_Dataset.csv	Dataset (CSV)	Python Output	The final cleaned master dataset containing all employee attributes plus calculated `Average_Working_Hours` for attrition modeling.
Employee Attrition Analysis	Presentation Deck	HTML Slide Deck	Visual executive presentation deck with key metrics, comparative charts, exit triggers, and strategic retention recommendations.

3. Key Analytical Findings (Leavers vs. Stayers)

Comparative analysis between employees who left the organization (Attrition = Yes) and those who remained (Attrition = No) revealed distinct drivers of turnover:

8.32 vs 7.58

AVG DAILY WORK
HOURS
(LEFT VS. STAYED)

₹61.6k vs

₹65.6k

AVG MONTHLY PAY
(LEFT VS. STAYED)

2.47 vs 2.77

ENVIRONMENT SAT.
(LEFT VS. STAYED / 4)

2.47 vs 2.78

JOB SATISFACTION
(LEFT VS. STAYED / 4)

Primary Exit Triggers Identified:

- **Workplace Burnout (Overtime):** Employees who exited worked nearly 1 additional hour per day on average (8.32 hrs/day vs. 7.58 hrs/day).
- **Compensation Disparity:** Exited employees received ~₹4,000 lower average monthly income compared to peers who stayed.
- **Lower Workplace & Job Satisfaction:** Lower environmental satisfaction (2.47) and job satisfaction (2.47) directly correlated with attrition risk.

4. Strategic Recommendations for HR Management

Actionable Retention Strategy:

1. **Cap Overtime & Enforce Work-Life Balance:** Implement strict overtime monitoring for teams averaging over 8 hours daily to reduce burnout.
2. **Realign Salary Benchmarks:** Conduct compensation reviews for high-risk roles where salaries fall below the ₹65,000 threshold.
3. **Enhance Workplace Environment:** Conduct quarterly pulse surveys to address manager-employee alignment and workplace culture issues.